Solar Mg appendix

Sun · Mg · downloadable appendix

Document	Solar Mg appendix
Subject	Solar photospheric abundance appendix · Mg
Author	Ryan Schmitt
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Publisher	The Exoplanet Codex · exoplanetcodex.org
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Web	https://exoplanetcodex.org/systems/sol/elements/mg/

Solar magnesium: 7.646 ± 0.095

Asplund, Amarsi & Grevesse 2021: 7.55 ± 0.03 · this measurement +0.096 dex

The three-dimensional non-LTE products apply published line-by-line corrections (Asplund et al. 2021) to our one-dimensional fits; they are not our own three-dimensional calculation. The K-band products are the first for magnesium. No second telluric correction exists in the K band, so their telluric error is scaled from the J band of the same night.

Our best HARPS product, chosen the same way as for iron: the Reference Grade analysis with the most complete treatment, then the most lines. HARPS is the instrument we use for the stars that follow, so the Sun is anchored on it. Products and engines are not averaged together.

VIS: 7.646 ± 0.095 · HARPS · VIS · Mg I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)

Highlighted Products

VIS

7.660 ± 0.089

Mg I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · Kitt Peak (Kurucz 2005) · Reference Grade

red-optical

7.663 ± 0.062

Mg I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

NIR

7.717 ± 0.124

Mg II · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

H

7.587 ± 0.088

Mg I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · CRIRES+ H · Reference Grade

K

7.507 ± 0.226

Mg I · Codex-graded lines · 1D NLTE (Gerber) · CRIRES+ · Codex Grade

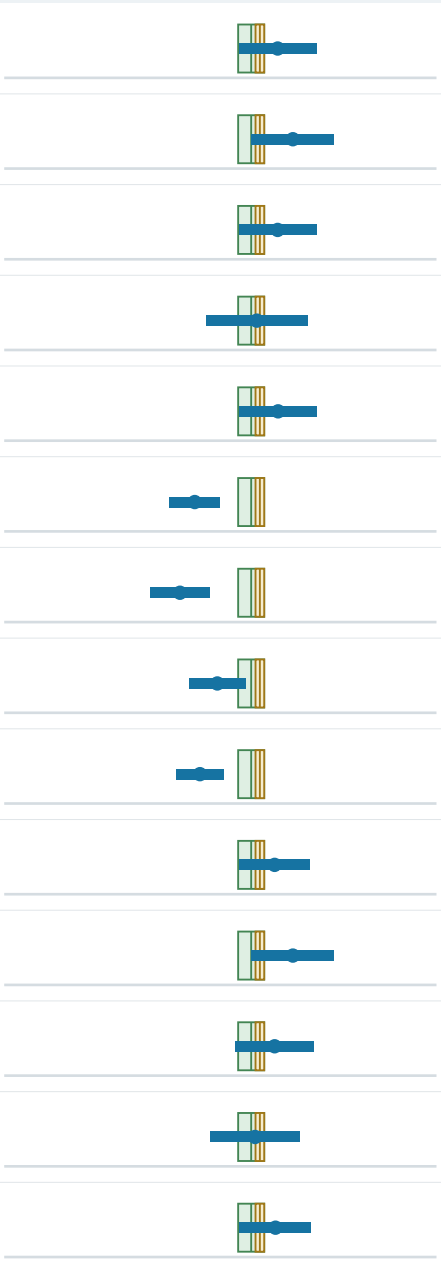
Line grades

Every product is measured on one of three grades of spectral lines and the grades are never mixed in one number.

Grade	What it means	This element
Reference Grade	lines from a published solar line set (Asplund et al. 2021 and similar), measured with the oscillator strengths the authors used	111 products -- Mg I · Asplund et al. 2021 lines; Mg II · Asplund et al. 2021 lines; SET-MG_BERGEMANN2017
Codex Grade	lines chosen by the Codex whose oscillator strength carries a published uncertainty (laboratory measurement, NIST accuracy class A-C, or the paper's own stated error), between 5% and 60% deep	88 products -- Mg I · Codex-graded lines; Mg II · Codex-graded lines
Deep Grade	the same oscillator-strength rule on lines deeper than 60% -- stronger lines, more sensitive to damping and the model atmosphere	20 products -- Mg I · Deep-graded lines

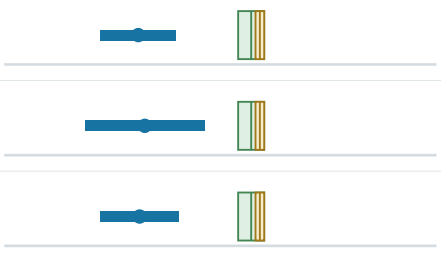
Error-bar Forest Plot

VIS — HARPS
A(Mg) dex; axis 6.98–7.98



Mg I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines	7.611 ±0.089
Mg I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 2 lines	7.646 ±0.095
Mg I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines	7.611 ±0.089
Mg I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 2 lines	7.563 ±0.117
Mg I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 2 lines	7.612 ±0.089
Mg I · Deep-graded lines · 1D LTE Deep Grade · 6 lines	7.420 ±0.058
Mg I · Deep-graded lines · 1D NLTE (Amarsi) Deep Grade · 4 lines	7.386 ±0.068
Mg I · Deep-graded lines · 1D NLTE (Gerber) Deep Grade · 6 lines	7.472 ±0.065
Mg I · Deep-graded lines · 1D LTE (Gerber) Deep Grade · 6 lines	7.432 ±0.054
Mg I · Codex-graded lines · 1D LTE Codex Grade · 3 lines	7.604 ±0.082
Mg I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 2 lines	7.646 ±0.095
Mg I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 3 lines	7.604 ±0.091
Mg I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 3 lines	7.559 ±0.103
Mg I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 3 lines	7.606 ±0.082

VIS — IAG FTS
A(Mg) dex; axis 6.98–7.98



SET-MG_BERGEMANN2017 · 1D LTE Reference Grade · 2 lines	7.290 ±0.087
SET-MG_BERGEMANN2017 · 1D NLTE (Amarsi) Reference Grade · 1 line	7.305 ±0.137
SET-MG_BERGEMANN2017 · 1D NLTE (Gerber) Reference Grade · 2 lines	7.293 ±0.091

VIS — IAG FTS
A(Mg) dex; axis 6.98–7.98

SET-MG_BERGEMANN2017 · 1D LTE (Gerber) Reference Grade · 2 lines		7.305 ±0.088
Mg I · Deep-graded lines · 1D LTE Deep Grade · 2 lines		7.290 ±0.087
Mg I · Deep-graded lines · 1D NLTE (Amarsi) Deep Grade · 1 line		7.305 ±0.137
Mg I · Deep-graded lines · 1D NLTE (Gerber) Deep Grade · 2 lines		7.293 ±0.091
Mg I · Deep-graded lines · 1D LTE (Gerber) Deep Grade · 2 lines		7.305 ±0.088

VIS — IAG FTS
A(Mg) dex; axis 6.98–7.98

Mg I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		7.623 ±0.088
Mg I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 2 lines		7.658 ±0.095
Mg I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		7.623 ±0.088
Mg I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 2 lines		7.595 ±0.104
Mg I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 2 lines		7.625 ±0.087
SET-MG_BERGEMANN2017 · 1D LTE Reference Grade · 4 lines		7.463 ±0.086
SET-MG_BERGEMANN2017 · 1D NLTE (Amarsi) Reference Grade · 3 lines		7.450 ±0.081
SET-MG_BERGEMANN2017 · 1D NLTE (Gerber) Reference Grade · 4 lines		7.492 ±0.409
SET-MG_BERGEMANN2017 · 1D LTE (Gerber) Reference Grade · 4 lines		7.474 ±0.074
Mg I · Deep-graded lines · 1D LTE Deep Grade · 4 lines		7.434 ±0.071
Mg I · Deep-graded lines · 1D NLTE (Amarsi) Deep Grade · 3 lines		7.398 ±0.057
Mg I · Deep-graded lines · 1D NLTE (Gerber) Deep Grade · 4 lines		7.479 ±0.468
Mg I · Deep-graded lines · 1D LTE (Gerber) Deep Grade · 4 lines		7.446 ±0.058

VIS — IAG FTS
A(Mg) dex; axis 6.98–7.98

Mg I · Codex-graded lines · 1D LTE Codex Grade · 3 lines		7.617 ±0.099
Mg I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 2 lines		7.658 ±0.095
Mg I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 3 lines		7.617 ±0.107
Mg I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 3 lines		7.578 ±0.106
Mg I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 3 lines		7.619 ±0.098

VIS — Kitt Peak (Kurucz 2005)
A(Mg) dex; axis 6.98–7.98

Mg I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		7.625 ±0.082
Mg I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 2 lines		7.660 ±0.089
Mg I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		7.625 ±0.082
Mg I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 2 lines		7.612 ±0.084
Mg I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 2 lines		7.627 ±0.081
Mg I · Deep-graded lines · 1D LTE Deep Grade · 6 lines		7.390 ±0.053
Mg I · Deep-graded lines · 1D NLTE (Amarsi) Deep Grade · 4 lines		7.384 ±0.064
Mg I · Deep-graded lines · 1D NLTE (Gerber) Deep Grade · 6 lines		7.414 ±0.117
Mg I · Deep-graded lines · 1D LTE (Gerber) Deep Grade · 6 lines		7.414 ±0.051
Mg I · Codex-graded lines · 1D LTE Codex Grade · 3 lines		7.615 ±0.107
Mg I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 2 lines		7.660 ±0.089
Mg I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 3 lines		7.615 ±0.116
Mg I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 3 lines		7.607 ±0.106

VIS — Kitt Peak (Kurucz 2005)
A(Mg) dex; axis 6.98–7.98

Mg I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 3 lines		7.617 ±0.106

VIS — Kitt Peak (1984, molecfit)
A(Mg) dex; axis 6.98–7.98

Mg I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		7.625 ±0.086
Mg I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 2 lines		7.660 ±0.093
Mg I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		7.625 ±0.086
Mg I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 2 lines		7.596 ±0.096
Mg I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 2 lines		7.627 ±0.086
Mg I · Deep-graded lines · 1D LTE Deep Grade · 6 lines		7.389 ±0.047
Mg I · Deep-graded lines · 1D NLTE (Amarsi) Deep Grade · 4 lines		7.384 ±0.061
Mg I · Deep-graded lines · 1D NLTE (Gerber) Deep Grade · 6 lines		7.418 ±0.120
Mg I · Deep-graded lines · 1D LTE (Gerber) Deep Grade · 6 lines		7.412 ±0.044
Mg I · Codex-graded lines · 1D LTE Codex Grade · 3 lines		7.617 ±0.110
Mg I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 2 lines		7.660 ±0.093
Mg I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 3 lines		7.617 ±0.120
Mg I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 3 lines		7.577 ±0.102
Mg I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 3 lines		7.619 ±0.109

VIS axis: 6.98 · 7.23 · 7.48 · 7.73 · 7.98 A(Mg) dex

red-optical — IAG FTS
A(Mg) dex; axis 7.28–7.85

Mg II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		7.522 ±0.152

red-optical — IAG FTS
A(Mg) dex; axis 7.28–7.85

Mg II · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 2 lines		7.505 ±0.146
Mg II · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		7.500 ±0.151
Mg II · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 2 lines		7.503 ±0.135
Mg II · Codex-graded lines · 1D LTE Codex Grade · 2 lines		7.522 ±0.152
Mg II · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 2 lines		7.505 ±0.146
Mg II · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines		7.500 ±0.151
Mg II · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 2 lines		7.503 ±0.135
Mg I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 3 lines		7.612 ±0.049
Mg I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 3 lines		7.661 ±0.063
Mg I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 3 lines		7.611 ±0.057
Mg I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 3 lines		7.607 ±0.049
Mg I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 3 lines		7.610 ±0.049
SET-MG_BERGEMANN2017 · 1D LTE Reference Grade · 1 line		7.561 ±0.256
SET-MG_BERGEMANN2017 · 1D NLTE (Gerber) Reference Grade · 1 line		7.490 ±0.063
SET-MG_BERGEMANN2017 · 1D LTE (Gerber) Reference Grade · 1 line		7.569 ±0.234
Mg I · Codex-graded lines · 1D LTE Codex Grade · 7 lines		7.591 ±0.059
Mg I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 3 lines		7.671 ±0.065
Mg I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 3 lines		7.618 ±0.060
Mg I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 7 lines		7.586 ±0.084

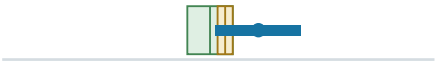
red-optical — IAG FTS
A(Mg) dex; axis 7.28–7.85

Mg I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 7 lines		7.589 ±0.059

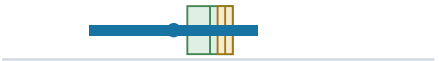
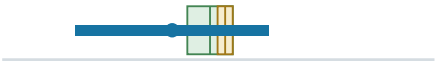
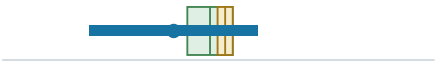
red-optical — Kitt Peak (Kurucz 2005)
A(Mg) dex; axis 7.28–7.85

Mg II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		7.523 ±0.141
Mg II · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 2 lines		7.505 ±0.134
Mg II · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		7.500 ±0.140
Mg II · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 2 lines		7.504 ±0.124
Mg II · Codex-graded lines · 1D LTE Codex Grade · 2 lines		7.523 ±0.141
Mg II · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 2 lines		7.505 ±0.134
Mg II · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines		7.500 ±0.140
Mg II · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 2 lines		7.504 ±0.124
Mg I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 3 lines		7.608 ±0.048
Mg I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 3 lines		7.657 ±0.062
Mg I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 3 lines		7.607 ±0.057
Mg I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 3 lines		7.598 ±0.053
Mg I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 3 lines		7.606 ±0.048
SET-MG_BERGEMANN2017 · 1D LTE Reference Grade · 1 line		7.557 ±0.250
SET-MG_BERGEMANN2017 · 1D NLTE (Gerber) Reference Grade · 1 line		7.488 ±0.074
SET-MG_BERGEMANN2017 · 1D LTE (Gerber) Reference Grade · 1 line		7.565 ±0.230
Mg I · Codex-graded lines · 1D LTE Codex Grade · 7 lines		7.584 ±0.049

red-optical — Kitt Peak (Kurucz 2005)
A(Mg) dex; axis 7.28–7.85

Mg I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 3 lines		7.660 ±0.062
Mg I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 3 lines		7.614 ±0.056
Mg I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 7 lines		7.566 ±0.052
Mg I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 7 lines		7.581 ±0.049

red-optical — Kitt Peak (1984, molecfit)
A(Mg) dex; axis 7.28–7.85

Mg II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		7.523 ±0.129
Mg II · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 2 lines		7.505 ±0.122
Mg II · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		7.500 ±0.128
Mg II · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 2 lines		7.502 ±0.111
Mg II · Codex-graded lines · 1D LTE Codex Grade · 2 lines		7.523 ±0.129
Mg II · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 2 lines		7.505 ±0.122
Mg II · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines		7.500 ±0.128
Mg II · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 2 lines		7.502 ±0.111
Mg I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 3 lines		7.614 ±0.048
Mg I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 3 lines		7.663 ±0.062
Mg I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 3 lines		7.613 ±0.056
Mg I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 3 lines		7.602 ±0.052
Mg I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 3 lines		7.612 ±0.048
SET-MG_BERGEMANN2017 · 1D LTE Reference Grade · 1 line		7.554 ±0.248

red-optical — Kitt Peak (1984, molecfit)
A(Mg) dex; axis 7.28–7.85

SET-MG_BERGEMANN2017 · 1D NLTE (Gerber) Reference Grade · 1 line		7.475 ±0.094
SET-MG_BERGEMANN2017 · 1D LTE (Gerber) Reference Grade · 1 line		7.563 ±0.226
Mg I · Codex-graded lines · 1D LTE Codex Grade · 7 lines		7.576 ±0.051
Mg I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 3 lines		7.666 ±0.061
Mg I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 3 lines		7.620 ±0.056
Mg I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 7 lines		7.568 ±0.054
Mg I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 7 lines		7.573 ±0.050

red-optical axis: 7.28 · 7.42 · 7.56 · 7.70 · 7.85 A(Mg) dex

NIR — CRIRES+ Y (full arm)
A(Mg) dex; axis 7.04–8.17

Mg II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		7.344 ±0.246
Mg II · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 1 line		7.321 ±0.249
Mg II · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 1 line		7.349 ±0.248
Mg II · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 1 line		7.306 ±0.240
Mg II · Codex-graded lines · 1D LTE Codex Grade · 1 line		7.344 ±0.246
Mg II · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 1 line		7.321 ±0.249
Mg II · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 1 line		7.349 ±0.248
Mg II · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 1 line		7.306 ±0.240
Mg I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 1 line		7.550 ±0.073
Mg I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 1 line		7.514 ±0.064

NIR — CRIRES+ Y (full arm)
A(Mg) dex; axis 7.04–8.17

Mg I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 2 lines		7.607 ±0.169
Mg I · Codex-graded lines · 1D LTE Codex Grade · 1 line		7.514 ±0.064
Mg I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 1 line		7.550 ±0.073
Mg I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 1 line		7.514 ±0.064
Mg I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 1 line		7.506 ±0.366
Mg I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 1 line		7.513 ±0.064

NIR — IAG FTS
A(Mg) dex; axis 7.04–8.17

Mg II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		7.725 ±0.179
Mg II · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 2 lines		7.703 ±0.179
Mg II · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		7.657 ±0.174
Mg II · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 2 lines		7.686 ±0.180
Mg II · Codex-graded lines · 1D LTE Codex Grade · 2 lines		7.725 ±0.179
Mg II · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 2 lines		7.703 ±0.179
Mg II · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines		7.657 ±0.174
Mg II · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 2 lines		7.686 ±0.180
Mg I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		7.649 ±0.160
Mg I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 1 line		7.640 ±0.183
Mg I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 1 line		7.648 ±0.161
Mg I · Codex-graded lines · 1D LTE Codex Grade · 5 lines		7.586 ±0.118

NIR — IAG FTS
A(Mg) dex; axis 7.04–8.17

Mg I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 5 lines		7.574 ±0.128
Mg I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 5 lines		7.584 ±0.119

NIR — Kitt Peak (Kurucz 2005)
A(Mg) dex; axis 7.04–8.17

Mg II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		7.879 ±0.256
Mg II · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 2 lines		7.857 ±0.255
Mg II · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		7.811 ±0.252
Mg II · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 2 lines		7.838 ±0.251
Mg II · Codex-graded lines · 1D LTE Codex Grade · 2 lines		7.879 ±0.256
Mg II · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 2 lines		7.857 ±0.255
Mg II · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines		7.811 ±0.252
Mg II · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 2 lines		7.838 ±0.251
Mg I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		7.661 ±0.118
Mg I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 1 line		7.646 ±0.212
Mg I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 1 line		7.661 ±0.118
Mg I · Codex-graded lines · 1D LTE Codex Grade · 4 lines		7.712 ±0.088
Mg I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 4 lines		7.687 ±0.097
Mg I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 4 lines		7.709 ±0.088

NIR — Kitt Peak (1984, molecfit)
A(Mg) dex; axis 7.04–8.17

Mg II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 4 lines		7.740 ±0.120
Mg II · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 4 lines		7.717 ±0.124
Mg II · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 4 lines		7.691 ±0.117
Mg II · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 4 lines		7.703 ±0.115
Mg II · Codex-graded lines · 1D LTE Codex Grade · 5 lines		7.637 ±0.104
Mg II · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 4 lines		7.646 ±0.123
Mg II · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 4 lines		7.638 ±0.115
Mg II · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 5 lines		7.602 ±0.096
Mg I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 1 line		7.608 ±0.072
Mg I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 1 line		7.572 ±0.063
Mg I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 2 lines		7.631 ±0.083
Mg I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 1 line		7.608 ±0.072
Mg I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 1 line		7.572 ±0.063
Mg I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 13 lines		7.535 ±0.138

NIR — CRIRES+ J
A(Mg) dex; axis 7.04–8.17

SET-MG_BERGEMANN2017 · 1D LTE Reference Grade · 2 lines		7.557 ±0.342
SET-MG_BERGEMANN2017 · 1D NLTE (Gerber) Reference Grade · 2 lines		7.511 ±0.208
SET-MG_BERGEMANN2017 · 1D LTE (Gerber) Reference Grade · 2 lines		7.564 ±0.334
Mg I · Codex-graded lines · 1D LTE Codex Grade · 7 lines		7.504 ±0.085

NIR — CRIRES+ J
A(Mg) dex; axis 7.04–8.17

Mg I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 6 lines		7.513 ±0.073
Mg I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 7 lines		7.504 ±0.092

NIR — CRIRES+ Y
A(Mg) dex; axis 7.04–8.17

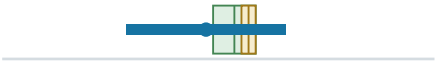
Mg I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		7.525 ±0.062
Mg I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 1 line		7.561 ±0.071
Mg I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 1 line		7.525 ±0.062
Mg I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 1 line		7.527 ±0.427
Mg I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 1 line		7.524 ±0.062

NIR axis: 7.04 · 7.32 · 7.60 · 7.88 · 8.17 A(Mg) dex

H — CRIRES+ H
A(Mg) dex; axis 7.22–7.83

Mg I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		7.535 ±0.079
Mg I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 1 line		7.587 ±0.088
Mg I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 1 line		7.534 ±0.084
Mg I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 1 line		7.527 ±0.187
Mg I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 1 line		7.538 ±0.079
SET-MG_BERGEMANN2017 · 1D LTE (Gerber) Reference Grade · 2 lines		7.561 ±0.241
Mg I · Codex-graded lines · 1D LTE Codex Grade · 8 lines		7.483 ±0.227
Mg I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 7 lines		7.569 ±0.203
Mg I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 9 lines		7.564 ±0.205

H — CRIRES+ H (full arm, telluric-corrected)
A(Mg) dex; axis 7.22–7.83

Mg I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		7.516 ±0.106
Mg I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 1 line		7.568 ±0.113
Mg I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 1 line		7.515 ±0.110
Mg I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 1 line		7.510 ±0.113
Mg I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 1 line		7.519 ±0.106
Mg I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 8 lines		7.380 ±0.127
Mg I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 9 lines		7.446 ±0.130

H axis: 7.22 · 7.37 · 7.53 · 7.68 · 7.83 A(Mg) dex

K — CRIRES+
A(Mg) dex; axis 7.25–7.76

Mg I · Codex-graded lines · 1D LTE Codex Grade · 11 lines		7.502 ±0.156
Mg I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 11 lines		7.507 ±0.226
Mg I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 11 lines		7.508 ±0.152

K axis: 7.25 · 7.38 · 7.51 · 7.64 · 7.76 A(Mg) dex

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Each bar is the total uncertainty -- the statistical and systematic terms combined in quadrature, as Asplund et al. 2021 and Lodders et al. 2025 quote it. The full error budget remains attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

How this was measured

Every magnesium value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the magnesium abundance as the free parameter. The synthesis includes the atomic lines and the molecular opacity in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; nothing is averaged across indicators or instruments. Where a 3D or NLTE correction is published for the lines, it is

applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit window, model atmosphere, telluric residual, and the spread between spectra, each measured by refitting with that one input changed. The headline at the top is the Asplund-grade product: a published line set, the most complete treatment, then the most lines.

Band	What was measured	Solar spectra	Treatments	Lines	Result A(Mg)	Largest uncertainty terms (dex)
VIS	Mg I · Deep-graded lines	HARPS, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	1–6	7.293–7.479 (1D-NLTE)	line strengths (gf) 0.04, fit / line-to-line scatter 0.04
VIS	Mg I · Codex-graded lines	HARPS, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE, 3D-NLTE	2–3	7.646–7.660 (3D-NLTE)	continuum placement 0.07, line strengths (gf) 0.05
VIS	Mg I · Asplund et al. 2021 lines	HARPS, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE, 3D-NLTE	2	7.646–7.660 (3D-NLTE)	continuum placement 0.07, line strengths (gf) 0.05
VIS	SET-MG_BERGEMANN2017	IAG FTS	1D-LTE, 1D-NLTE	1–4	7.293–7.492 (1D-NLTE)	fit window 0.41, line strengths (gf) 0.04
red-optical	Mg I · Codex-graded lines; Mg II · Codex-graded lines	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE, 3D-NLTE	2–7	7.505–7.671 (3D-NLTE)	line strengths (gf) 0.04, NLTE / 3D correction 0.03
red-optical	Mg I · Asplund et al. 2021 lines; Mg II · Asplund et al. 2021 lines	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE, 3D-NLTE	2–3	7.505–7.663 (3D-NLTE)	line strengths (gf) 0.04, NLTE / 3D correction 0.03
red-optical	SET-MG_BERGEMANN2017	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	1	7.475–7.490 (1D-NLTE)	fit window 0.04, line strengths (gf) 0.03
NIR	Mg I · Codex-graded lines; Mg II · Codex-graded lines	CRIRES+ J, CRIRES+ Y (full arm), IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE, 3D-NLTE	1–13	7.321–7.857 (3D-NLTE)	fit / line-to-line scatter 0.10, telluric residual 0.06
NIR	Mg I · Asplund et al. 2021 lines; Mg II · Asplund et al. 2021 lines	CRIRES+ Y, CRIRES+ Y (full arm), IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE, 3D-NLTE	1–4	7.321–7.857 (3D-NLTE)	fit / line-to-line scatter 0.10, telluric residual 0.06
NIR	SET-MG_BERGEMANN2017	CRIRES+ J	1D-LTE, 1D-NLTE	2	7.511 (1D-NLTE)	fit window 0.18, fit / line-to-line scatter 0.09
H	Mg I · Codex-graded lines	CRIRES+ H, CRIRES+ H (full arm, telluric-corrected)	1D-LTE, 1D-NLTE	7–9	7.380–7.569 (1D-NLTE)	telluric residual 0.11, fit / line-to-line scatter 0.04

Band	What was measured	Solar spectra	Treatments	Lines	Result A(Mg)	Largest uncertainty terms (dex)
H	Mg I · Asplund et al. 2021 lines	CRIRES+ H, CRIRES+ H (full arm, telluric-corrected)	1D-LTE, 1D-NLTE, 3D-NLTE	1	7.568–7.587 (3D-NLTE)	line strengths (gf) 0.06, fit / line-to-line scatter 0.04
H	SET-MG_BERGEMANN2017	CRIRES+ H	1D-LTE	2	7.561 (1D-LTE)	fit / line-to-line scatter 0.21, telluric residual 0.11
K	Mg I · Codex-graded lines	CRIRES+	1D-LTE, 1D-NLTE	11	7.507 (1D-NLTE)	telluric residual 0.22, fit / line-to-line scatter 0.05

Excluded lines

These lines are left out of the Sun's measurement. They stay in the line lists: a cooler or more evolved star may measure them well. Blended lines are not excluded -- the spectrum synthesis models the blend. 8 lines.

Saturated: on the flat part of the curve of growth, so the line barely responds to the abundance (reduced width log W/λ above -4.9).

Species	Wavelength (Å, air)	Why it is excluded
Mg I	3091.064	saturated
Mg I	3092.985	saturated
Mg I	3096.890	saturated
Mg I	3332.146	saturated
Mg I	3336.674	saturated
Mg I	3627.628	saturated
Mg I	15748.988	atomic_blend
Mg I	24566.296	atomic_blend

Solar magnesium context

Magnesium is measured separately from neutral Mg I and ionized Mg II. Asplund et al. 2021 use eight Mg I and six Mg II features, average the lines within each ion, then weight the two ion means by their total uncertainties. Both ion results are required for the combined abundance.

Published line sets · gf uncertainty model · Methodology

Download the evidence

Download PDF — Solar Mg appendix

Source feed — includes quarantined fit outputs, not adopted abundances

Complete product visibility audit

Shared website/PDF report

Reproducibility manifest

Complete references and product-to-source bindings

Reproducibility

Generated from the pinned science snapshot. Source commit 69569fe3c4a1345b35923cb772cb3b06be4ab2d5
Feed Mg.json v1.335 · updated 2026-10-28T21:28:27Z
Generated 2026-10-28T15:35:12-06:00 (source commit timestamp)
Feed SHA-256 1ec795f457be0118c48c6108a72384016a1ab5244a7cc4ac0385d43f01204046
Generator scripts/generate_cno_publication.py v1.0.0

Pinned scientific source

References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

Solar abundance and indicator analyses. Solar Mg comparison literature; not an adopted Codex abundance. Sources: [1], [2].

Solar abundance and indicator analyses. Published line set MG_AGSS21 (the line selection). Sources: [1].

Atomic data and gf authorities. Mg II oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [1].

Model and grid sources. Feed treatment: 1D-LTE synthesis (Turbospectrum); no NLTE/3D correction claimed. Sources: [3].

Model and grid sources. Cited solar abundance differences from the published model columns, applied additively to our LTE fits; not a new in-transfer 3D calculation. Baseline atmosphere: A_marcs_lte Sources: [1].

Model and grid sources. Feed treatment: synth-1D-LTE-gerber; Turbospectrum on the registered Gerber deck. The product retains its own atmosphere and ion identity. Sources: [3].

Atomic data and gf authorities. Mg I oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [1].

Model and grid sources. Feed treatment: ENGINE-B-NLTE; Turbospectrum on the registered Gerber deck. The product retains its own atmosphere and ion identity. Sources: [3].

Solar abundance and indicator analyses. Published line set MG_BERGEMANN2017 (the line selection). Sources: [5].

Model and grid sources. Feed treatment: 1D NLTE, per-line correction from the Amarsi 2020 GALAH departure grid via PySME (data/nlte_grids/Mg_Amarsi2020_PySME.csv). Sources: [7].

Product-to-source index

NIR · CRIRES+ Y (full arm) · Mg II · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3] Mg-products-d816dfae18d8

NIR · CRIRES+ Y (full arm) · Mg II · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [2] Mg-products-6d4fdfbb0d6b

NIR · CRIRES+ Y (full arm) · Mg II · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2] Mg-products-afd7d0ed8741

NIR · CRIRES+ Y (full arm) · Mg II · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [3] Mg-products-da69d3d9451a

NIR · CRIRES+ Y (full arm) · Mg II · Codex-graded lines · 1D LTE · [1], [2], [3] Mg-products-344fb65a8b45

NIR · CRIRES+ Y (full arm) · Mg II · Codex-graded lines · 3D NLTE (Amarsi) · [1], [2] Mg-products-3ac4e3f1be12

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H · CRIRES+ H · Mg I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [6], [4], [3] Mg-products-e91ea5062469

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H · CRIRES+ H (full arm, telluric-corrected) · Mg I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2] Mg-products-81d206179460

H · CRIRES+ H (full arm, telluric-corrected) · Mg I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [3] Mg-products-10597f41c021

H · CRIRES+ H (full arm, telluric-corrected) · Mg I · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [3] Mg-products-ce8db0d66041

H · CRIRES+ H (full arm, telluric-corrected) · Mg I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [6], [4], [5], [3] Mg-products-d8bb5b97cc7e

H · CRIRES+ H (full arm, telluric-corrected) · Mg I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [6], [4], [5], [3] Mg-products-1909b40189b8

K · CRIRES+ · Mg I · Codex-graded lines · 1D LTE · [1], [2], [4], [3] Mg-products-99de8cd7cbcc

K · CRIRES+ · Mg I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [4], [3] Mg-products-3815e328ddc4

K · CRIRES+ · Mg I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [4], [3] Mg-products-275eece75f99

NIR · CRIRES+ J · SET-MG_BERGEMANN2017 · 1D LTE · [1], [2], [5], [6], [3] Mg-products-f8f393f28370

NIR · CRIRES+ J · SET-MG_BERGEMANN2017 · 1D NLTE (Gerber) · [1], [2], [5], [6], [3] Mg-products-84060533653f

NIR · CRIRES+ J · SET-MG_BERGEMANN2017 · 1D LTE (Gerber) · [1], [2], [5], [6], [3] Mg-products-07cae91a52f9

NIR · CRIRES+ J · Mg I · Codex-graded lines · 1D LTE · [1], [2], [4], [6], [3] Mg-products-c03040e070f0

NIR · CRIRES+ J · Mg I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [6], [4], [3] Mg-products-6c343e995cfa

NIR · CRIRES+ J · Mg I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [4], [6], [3] Mg-products-da24809844c2

NIR · CRIRES+ Y · Mg I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3] Mg-products-49eb24c3495c

NIR · CRIRES+ Y · Mg I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [2] Mg-products-21d94165adac

NIR · CRIRES+ Y · Mg I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2] Mg-products-a35633649cd9

NIR · CRIRES+ Y · Mg I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [3] Mg-products-39039bd84318

NIR · CRIRES+ Y · Mg I · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [3] Mg-products-e02bda0de567

NIR · CRIRES+ Y (full arm) · Mg I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [2] Mg-products-3e2ab2c305c1

NIR · CRIRES+ Y (full arm) · Mg I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2] Mg-products-35b746de3e52

NIR · CRIRES+ Y (full arm) · Mg I · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [3] Mg-products-67679071c2b6

NIR · CRIRES+ Y (full arm) · Mg I · Codex-graded lines · 1D LTE · [1], [2], [3] Mg-products-476cbbf482f3

NIR · CRIRES+ Y (full arm) · Mg I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [2] Mg-products-c66ff6d91045

NIR · CRIRES+ Y (full arm) · Mg I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2] Mg-products-2cbaaaf5d30e

NIR · CRIRES+ Y (full arm) · Mg I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [3] Mg-products-048ff49242dd

NIR · CRIRES+ Y (full arm) · Mg I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [3] Mg-products-a2f38d3445a1

NIR · IAG FTS · Mg I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3] Mg-products-4d4e9a277be8

NIR · IAG FTS · Mg I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [3] Mg-products-1f2cdb8ac458

NIR · IAG FTS · Mg I · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [3] Mg-products-90c5d687a59b

NIR · IAG FTS · Mg I · Codex-graded lines · 1D LTE · [1], [2], [4], [6], [3] Mg-products-ab59ff5768fd

NIR · IAG FTS · Mg I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [4], [6], [3] Mg-products-8e5ad708074f

NIR · IAG FTS · Mg I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [4], [6], [3] Mg-products-ea0ee54ea759

NIR · Kitt Peak (Kurucz 2005) · Mg I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3] Mg-products-56b9cee76fcd

NIR · Kitt Peak (Kurucz 2005) · Mg I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [3] Mg-products-9100e741695c

NIR · Kitt Peak (Kurucz 2005) · Mg I · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [3] Mg-products-0bd811bed24e

NIR · Kitt Peak (Kurucz 2005) · Mg I · Codex-graded lines · 1D LTE · [1], [2], [6], [4], [3] Mg-products-643f43b2cc04

NIR · Kitt Peak (Kurucz 2005) · Mg I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [6], [4], [3] Mg-products-7a3dcd7b98aa

NIR · Kitt Peak (Kurucz 2005) · Mg I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [6], [4], [3] Mg-products-9d0180e23158

NIR · Kitt Peak (1984, molecfit) · Mg I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [2] Mg-products-e8490b8269f6

NIR · Kitt Peak (1984, molecfit) · Mg I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2] Mg-products-9edf762d80c4

NIR · Kitt Peak (1984, molecfit) · Mg I · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [3] Mg-products-48a1b81b0b57

NIR · Kitt Peak (1984, molecfit) · Mg I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [2] Mg-products-3f2e7d21bb9d

NIR · Kitt Peak (1984, molecfit) · Mg I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2] Mg-products-f631a82c728e

NIR · Kitt Peak (1984, molecfit) · Mg I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [6], [4], [3] Mg-products-f2232bce3c27

VIS · HARPS · Mg I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3] Mg-products-8771ce27433e

VIS · HARPS · Mg I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [2] Mg-products-625d514c9752

VIS · HARPS · Mg I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2] Mg-products-010b8e83ddb0

VIS · HARPS · Mg I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [3] Mg-products-2c00eba7fd54

VIS · HARPS · Mg I · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [3] Mg-products-fff8b029531a

VIS · HARPS · Mg I · Deep-graded lines · 1D LTE · [1], [2], [5], [6], [3] Mg-products-9e9ca5224ad4

VIS · HARPS · Mg I · Deep-graded lines · 1D NLTE (Amarsi) · [1], [2], [5], [6], [7] Mg-products-26ad0da45e60

VIS · HARPS · Mg I · Deep-graded lines · 1D NLTE (Gerber) · [1], [2], [5], [6], [3] Mg-products-281471766b3f

VIS · HARPS · Mg I · Deep-graded lines · 1D LTE (Gerber) · [1], [2], [5], [6], [3] Mg-products-80e3352427f0

VIS · HARPS · Mg I · Codex-graded lines · 1D LTE · [1], [2], [5], [3] Mg-products-dec40986e5ce

VIS · HARPS · Mg I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [2] Mg-products-3677c36e9e6b

VIS · HARPS · Mg I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [5] Mg-products-991705b37a93

VIS · HARPS · Mg I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [5], [3] Mg-products-51aca9f1e3aa

VIS · HARPS · Mg I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [5], [3] Mg-products-c0b3b7abc270

VIS · IAG FTS · SET-MG_BERGEMANN2017 · 1D LTE · [1], [2], [5], [3] Mg-products-4696b55aa616

VIS · IAG FTS · SET-MG_BERGEMANN2017 · 1D NLTE (Amarsi) · [1], [2], [5], [7] Mg-products-3ae879be8ac0

VIS · IAG FTS · SET-MG_BERGEMANN2017 · 1D NLTE (Gerber) · [1], [2], [5], [3] Mg-products-8eefdf45ae68

VIS · IAG FTS · SET-MG_BERGEMANN2017 · 1D LTE (Gerber) · [1], [2], [5], [3] Mg-products-a2508c413318

VIS · IAG FTS · Mg I · Deep-graded lines · 1D LTE · [1], [2], [5], [3] Mg-products-9df424db2ab1

VIS · IAG FTS · Mg I · Deep-graded lines · 1D NLTE (Amarsi) · [1], [2], [5], [7] Mg-products-94a25c354585

VIS · IAG FTS · Mg I · Deep-graded lines · 1D NLTE (Gerber) · [1], [2], [5], [3] Mg-products-4e5664d68ee7

VIS · IAG FTS · Mg I · Deep-graded lines · 1D LTE (Gerber) · [1], [2], [5], [3] Mg-products-da70ed9c96c4

VIS · IAG FTS · Mg I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3] Mg-products-a90ce2eb5c46

VIS · IAG FTS · Mg I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [2] Mg-products-b83545aa77c6

VIS · IAG FTS · Mg I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2] Mg-products-eecd9ac5b88c

VIS · IAG FTS · Mg I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [3] Mg-products-3681f9e08e93

VIS · IAG FTS · Mg I · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [3] Mg-products-24921a58783a

VIS · IAG FTS · SET-MG_BERGEMANN2017 · 1D LTE · [1], [2], [5], [6], [3] Mg-products-0ac266ce7380

VIS · IAG FTS · SET-MG_BERGEMANN2017 · 1D NLTE (Amarsi) · [1], [2], [5], [6], [7] Mg-products-840ae3c1bf82

VIS · IAG FTS · SET-MG_BERGEMANN2017 · 1D NLTE (Gerber) · [1], [2], [5], [6], [3] Mg-products-89c2d606613c

VIS · IAG FTS · SET-MG_BERGEMANN2017 · 1D LTE (Gerber) · [1], [2], [5], [6], [3] Mg-products-3d7238ba23ec

VIS · IAG FTS · Mg I · Deep-graded lines · 1D LTE · [1], [2], [6], [3] Mg-products-0e5af60d768f

VIS · IAG FTS · Mg I · Deep-graded lines · 1D NLTE (Amarsi) · [1], [2], [6], [7] Mg-products-41e21d2104ff

VIS · IAG FTS · Mg I · Deep-graded lines · 1D NLTE (Gerber) · [1], [2], [6], [3] Mg-products-6a2948b06f72

VIS · IAG FTS · Mg I · Deep-graded lines · 1D LTE (Gerber) · [1], [2], [6], [3] Mg-products-3ed29fecbaa1

VIS · IAG FTS · Mg I · Codex-graded lines · 1D LTE · [1], [2], [5], [3] Mg-products-a297f484f699

VIS · IAG FTS · Mg I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [2] Mg-products-17697e97cfdd

VIS · IAG FTS · Mg I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [5] Mg-products-13a52c28e19c

VIS · IAG FTS · Mg I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [5], [3] Mg-products-1931bc8ea12f

VIS · IAG FTS · Mg I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [5], [3] Mg-products-dc0225351b02

VIS · Kitt Peak (Kurucz 2005) · Mg I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3] Mg-products-6ced790f3d32

VIS · Kitt Peak (Kurucz 2005) · Mg I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [2] Mg-products-a403d95e022f

VIS · Kitt Peak (Kurucz 2005) · Mg I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2] Mg-products-987d5a975f8f

VIS · Kitt Peak (Kurucz 2005) · Mg I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [3] Mg-products-8bdd15dc1b51

VIS · Kitt Peak (Kurucz 2005) · Mg I · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [3] Mg-products-cf5a684ca2b4

VIS · Kitt Peak (Kurucz 2005) · Mg I · Deep-graded lines · 1D LTE · [1], [2], [5], [6], [3] Mg-products-b717c6cb3b4a

VIS · Kitt Peak (Kurucz 2005) · Mg I · Deep-graded lines · 1D NLTE (Amarsi) · [1], [2], [5], [6], [7] Mg-products-78a1d90b2e01

VIS · Kitt Peak (Kurucz 2005) · Mg I · Deep-graded lines · 1D NLTE (Gerber) · [1], [2], [5], [6], [3] Mg-products-10eb33d4fcd7

VIS · Kitt Peak (Kurucz 2005) · Mg I · Deep-graded lines · 1D LTE (Gerber) · [1], [2], [5], [6], [3] Mg-products-06a3b1c32947

VIS · Kitt Peak (Kurucz 2005) · Mg I · Codex-graded lines · 1D LTE · [1], [2], [5], [3] Mg-products-67fab9051db0

VIS · Kitt Peak (Kurucz 2005) · Mg I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [2] Mg-products-48dfc9affc99

VIS · Kitt Peak (Kurucz 2005) · Mg I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [5] Mg-products-fe95ac4bf5e4

VIS · Kitt Peak (Kurucz 2005) · Mg I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [5], [3] Mg-products-ee5c33bacacc

VIS · Kitt Peak (Kurucz 2005) · Mg I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [5], [3] Mg-products-ac53fbfb3f32

VIS · Kitt Peak (1984, molecfit) · Mg I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3] Mg-products-66cd3d2de3e5

VIS · Kitt Peak (1984, molecfit) · Mg I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [2] Mg-products-84c3ea38cb28

VIS · Kitt Peak (1984, molecfit) · Mg I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2] Mg-products-2ef30b20e600

VIS · Kitt Peak (1984, molecfit) · Mg I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [3] Mg-products-b5fe4d2f6516

VIS · Kitt Peak (1984, molecfit) · Mg I · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [3] Mg-products-d2a9ccbc5c2c

VIS · Kitt Peak (1984, molecfit) · Mg I · Deep-graded lines · 1D LTE · [1], [2], [5], [6], [3] Mg-products-6c93718b3f10

VIS · Kitt Peak (1984, molecfit) · Mg I · Deep-graded lines · 1D NLTE (Amarsi) · [1], [2], [5], [6], [7] Mg-products-558762314eec

VIS · Kitt Peak (1984, molecfit) · Mg I · Deep-graded lines · 1D NLTE (Gerber) · [1], [2], [5], [6], [3] Mg-products-d5e1d1327ac2

VIS · Kitt Peak (1984, molecfit) · Mg I · Deep-graded lines · 1D LTE (Gerber) · [1], [2], [5], [6], [3] Mg-products-498109e2200f

VIS · Kitt Peak (1984, molecfit) · Mg I · Codex-graded lines · 1D LTE · [1], [2], [5], [3] Mg-products-570f1bbcc507

VIS · Kitt Peak (1984, molecfit) · Mg I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [2] Mg-products-d075470f76f2

VIS · Kitt Peak (1984, molecfit) · Mg I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [5] Mg-products-69614e28dc32

VIS · Kitt Peak (1984, molecfit) · Mg I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [5], [3] Mg-products-a87e48b3c5ac

VIS · Kitt Peak (1984, molecfit) · Mg I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [5], [3] Mg-products-0a2c596ab198

red-optical · IAG FTS · Mg I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3] Mg-products-d7b0d2bf7001

red-optical · IAG FTS · Mg I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [2] Mg-products-d11440423950

red-optical · IAG FTS · Mg I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2] Mg-products-ebf245d08b35

red-optical · IAG FTS · Mg I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [3] Mg-products-862c4b99bb91

red-optical · IAG FTS · Mg I · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [3] Mg-products-15bbec7c657a

red-optical · IAG FTS · SET-MG_BERGEMANN2017 · 1D LTE · [1], [2], [5], [6], [3] Mg-products-7fc0876ab604

red-optical · IAG FTS · SET-MG_BERGEMANN2017 · 1D NLTE (Gerber) · [1], [2], [5], [6], [3] Mg-products-67c7e96e6be4

red-optical · IAG FTS · SET-MG_BERGEMANN2017 · 1D LTE (Gerber) · [1], [2], [5], [6], [3] Mg-products-21a72e80ad53

red-optical · IAG FTS · Mg I · Codex-graded lines · 1D LTE · [1], [2], [6], [3] Mg-products-0f5ea4b49036

red-optical · IAG FTS · Mg I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [2] Mg-products-42dc4354bf76

red-optical · IAG FTS · Mg I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2] Mg-products-a22d4e3ed1e0

red-optical · IAG FTS · Mg I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [6], [3] Mg-products-cef83605070c

red-optical · IAG FTS · Mg I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [6], [3] Mg-products-f4d6a6e9d117

red-optical · Kitt Peak (Kurucz 2005) · Mg I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3]
Mg-products-c19d5c74f97f

red-optical · Kitt Peak (Kurucz 2005) · Mg I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [2]
Mg-products-0cfadc7deb04

red-optical · Kitt Peak (Kurucz 2005) · Mg I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2]
Mg-products-86ba51bd3894

red-optical · Kitt Peak (Kurucz 2005) · Mg I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [3]
Mg-products-f00a96d2f9fe

red-optical · Kitt Peak (Kurucz 2005) · Mg I · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [3]
Mg-products-359323286f0a

red-optical · Kitt Peak (Kurucz 2005) · SET-MG_BERGEMANN2017 · 1D LTE · [1], [2], [5], [6], [3]
Mg-products-f8caee619e5e

red-optical · Kitt Peak (Kurucz 2005) · SET-MG_BERGEMANN2017 · 1D NLTE (Gerber) · [1], [2], [5], [6], [3]
Mg-products-0cc66451e2af

red-optical · Kitt Peak (Kurucz 2005) · SET-MG_BERGEMANN2017 · 1D LTE (Gerber) · [1], [2], [5], [6], [3]
Mg-products-5b6308a762c9

red-optical · Kitt Peak (Kurucz 2005) · Mg I · Codex-graded lines · 1D LTE · [1], [2], [6], [3]
Mg-products-a1f7c896d128

red-optical · Kitt Peak (Kurucz 2005) · Mg I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [2]
Mg-products-d20bc465314f

red-optical · Kitt Peak (Kurucz 2005) · Mg I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2]
Mg-products-1600201113a1

red-optical · Kitt Peak (Kurucz 2005) · Mg I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [6], [3]
Mg-products-325f5b9389f2

red-optical · Kitt Peak (Kurucz 2005) · Mg I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [6], [3]
Mg-products-50d47863ea8a

red-optical · Kitt Peak (1984, molecfit) · Mg I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3]
Mg-products-3fa2a5636fd3

red-optical · Kitt Peak (1984, molecfit) · Mg I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [2]
Mg-products-16da74be3498

red-optical · Kitt Peak (1984, molecfit) · Mg I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2]
Mg-products-3f1ae6a3fa00

red-optical · Kitt Peak (1984, molecfit) · Mg I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [3]
Mg-products-3b0126651cba

red-optical · Kitt Peak (1984, molecfit) · Mg I · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [3]
Mg-products-7c8d668cbddb

red-optical · Kitt Peak (1984, molecfit) · SET-MG_BERGEMANN2017 · 1D LTE · [1], [2], [5], [6], [3]
Mg-products-212e7efa6132

red-optical · Kitt Peak (1984, molecfit) · SET-MG_BERGEMANN2017 · 1D NLTE (Gerber) · [1], [2], [5], [6], [3]
Mg-products-21d9081e8ffb

red-optical · Kitt Peak (1984, molecfit) · SET-MG_BERGEMANN2017 · 1D LTE (Gerber) · [1], [2], [5], [6], [3]
Mg-products-d3a361170ac2

red-optical · Kitt Peak (1984, molecfit) · Mg I · Codex-graded lines · 1D LTE · [1], [2], [6], [3] Mg-products-4020bf635822

red-optical · Kitt Peak (1984, molecfit) · Mg I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [2] Mg-products-251020a2f5a1

red-optical · Kitt Peak (1984, molecfit) · Mg I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2] Mg-products-01e43654de61

red-optical · Kitt Peak (1984, molecfit) · Mg I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [6], [3] Mg-products-45eec9675889

red-optical · Kitt Peak (1984, molecfit) · Mg I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [6], [3] Mg-products-75303c7199f4

VIS · Kitt Peak (Kurucz 2005) · SET-MG_BERGEMANN2017 · 1D LTE (Gerber) · [1], [2], [5], [6], [3] Mg-quarantine-b52d72dd9990

VIS · Kitt Peak (Kurucz 2005) · SET-MG_BERGEMANN2017 · 1D NLTE (Gerber) · [1], [2], [5], [6], [3] Mg-quarantine-8643f7bb20af

VIS · Kitt Peak (Kurucz 2005) · SET-MG_BERGEMANN2017 · 1D LTE · [1], [2], [5], [6], [3] Mg-quarantine-1a1af53a4420

VIS · Kitt Peak (Kurucz 2005) · SET-MG_BERGEMANN2017 · 1D NLTE (Amarsi) · [1], [2], [5], [6], [7] Mg-quarantine-5afeb6e574ca

VIS · Kitt Peak (1984, molecfit) · SET-MG_BERGEMANN2017 · 1D LTE (Gerber) · [1], [2], [5], [6], [3] Mg-quarantine-36b43478e36a

VIS · Kitt Peak (1984, molecfit) · SET-MG_BERGEMANN2017 · 1D NLTE (Gerber) · [1], [2], [5], [6], [3] Mg-quarantine-93d9594958c1

VIS · Kitt Peak (1984, molecfit) · SET-MG_BERGEMANN2017 · 1D LTE · [1], [2], [5], [6], [3] Mg-quarantine-4e9f22e2a3b0

VIS · Kitt Peak (1984, molecfit) · SET-MG_BERGEMANN2017 · 1D NLTE (Amarsi) · [1], [2], [5], [6], [7] Mg-quarantine-8edced6ff6e1

VIS · HARPS · SET-MG_BERGEMANN2017 · 1D LTE (Gerber) · [1], [2], [5], [6], [3] Mg-quarantine-f25842530f9a

VIS · HARPS · SET-MG_BERGEMANN2017 · 1D NLTE (Gerber) · [1], [2], [5], [6], [3] Mg-quarantine-b3a8eca5cdb8

VIS · HARPS · SET-MG_BERGEMANN2017 · 1D LTE · [1], [2], [5], [6], [3] Mg-quarantine-3b5e57d664fb

VIS · HARPS · SET-MG_BERGEMANN2017 · 1D NLTE (Amarsi) · [1], [2], [5], [6], [7] Mg-quarantine-182e08af1233

NIR · Kitt Peak (1984, molecfit) · SET-MG_BERGEMANN2017 · 1D LTE (Gerber) · [1], [2], [5], [6], [3] Mg-quarantine-74110dd170d0

NIR · Kitt Peak (1984, molecfit) · Mg I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [6], [4], [3] Mg-quarantine-ed8a22bc135b

NIR · Kitt Peak (1984, molecfit) · Mg I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [3] Mg-quarantine-da22d8adab31

NIR · Kitt Peak (1984, molecfit) · SET-MG_BERGEMANN2017 · 1D NLTE (Gerber) · [1], [2], [5], [6], [3] Mg-quarantine-c0fc0f5dd738

NIR · Kitt Peak (1984, molecfit) · Mg I · Codex-graded lines · 1D LTE · [1], [2], [6], [4], [3] Mg-quarantine-9b9fe8bbfe3e

NIR · Kitt Peak (1984, molecfit) · Mg I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3] Mg-quarantine-adb3dfe1171e

NIR · Kitt Peak (1984, molecfit) · SET-MG_BERGEMANN2017 · 1D LTE · [1], [2], [5], [6], [3] Mg-quarantine-e06e523c799c

NIR · CRIRES+ Y (full arm) · Mg I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [3] Mg-quarantine-bdcc0243b7a9

NIR · CRIRES+ Y (full arm) · Mg I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3] Mg-quarantine-60c9f0540677

H · CRIRES+ H · SET-MG_BERGEMANN2017 · 1D NLTE (Gerber) · [1], [2], [5], [6], [3] Mg-quarantine-43a66451f55f

H · CRIRES+ H (full arm, telluric-corrected) · SET-MG_BERGEMANN2017 · 1D LTE (Gerber) · [1], [2], [5], [6], [3] Mg-quarantine-cfec46cea0cf

H · CRIRES+ H (full arm, telluric-corrected) · SET-MG_BERGEMANN2017 · 1D NLTE (Gerber) · [1], [2], [5], [6], [3] Mg-quarantine-5ed77bdd362a

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